CSA5112 - Cryptography and Network Security

12. RSA Algorithm

Aim: To implement the RSA algorithm for public key encryption and decryption of a plaintext message.

Algorithm:

1. Start.
2. Import the required RSA libraries.
3. Generate an RSA public key and private key.
4. Read the plaintext from the user.
5. Encrypt the plaintext using the public key.
6. Display the encrypted ciphertext.
7. Decrypt the ciphertext using the private key.
8. Display the original plaintext.
9. Stop.

Code:

```
from Crypto.PublicKey import RSA

from Crypto.Cipher import PKCS1_OAEP

import base64

# Generate RSA key pair

key = RSA.generate(2048)

public_key = key.publickey()

private_key = key

encrypt_cipher = PKCS1_OAEP.new(public_key)

decrypt_cipher = PKCS1_OAEP.new(private_key)
```

Read plaintext

plaintext = input("Enter Plain Text: ")

Encrypt

encrypted = encrypt_cipher.encrypt(plaintext.encode())

print("\nEncrypted Text:")

print(base64.b64encode(encrypted).decode())

Decrypt

decrypted = decrypt_cipher.decrypt(encrypted)

print("\nDecrypted Text:")

print(decrypted.decode())

Sample Input:

```
Enter Plain Text: HELLO
```

Sample Output:

```
Enter Plain Text:
HELLO WORLD

Encrypted Text:
Q4Nw2hW7eWm2d4P8mM6bLq5rK9VxY7dT1...
(Base64 ciphertext - will be different every time)

Decrypted Text:
HELLO WORLD
```

Result:

The RSA algorithm was successfully implemented. The plaintext was encrypted using the RSA public key and successfully decrypted using the corresponding private key.